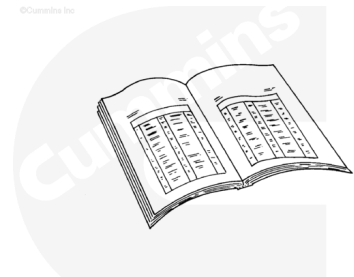


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

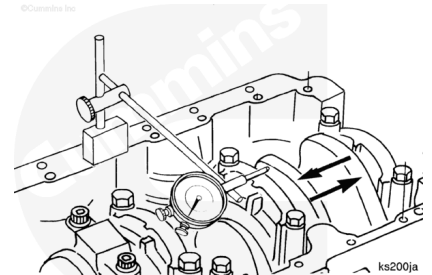
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

- Disconnect the batteries. Refer to the equipment manufacture service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/216/216-007-037.html>)
- Remove the lubricating oil pan and gasket. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/216/216-007-025-tr.html>)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/220/220-007-035-tr.html>)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.

Initial Check

Note : The dimensions of the thrust bearing and crankshaft journal determine end clearance.

Measure the crankshaft end clearance using Cummins® dial indicator, Part Number 3824564, and magnetic base, Part Number 3377399.

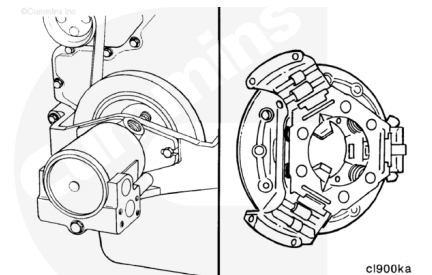


Crankshaft End Clearance		
mm		in
0.102	MIN	0.004
0.432	MAX	0.017

If the crankshaft end clearance is **not** within specification, make sure to inspect the crankshaft and thrust bearing surfaces for damage.

A common cause for increased crankshaft end clearance and thrust bearing damage is increased end-loading of the engine. The increased end-loading can be the result of driven units at the front or rear of the engine being:

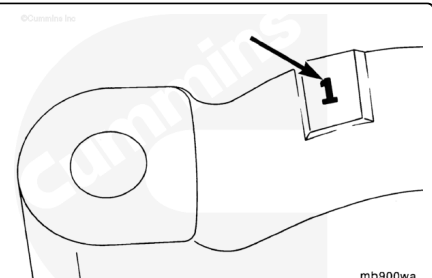
- Incorrectly installed
- Incorrectly adjusted
- Incorrectly matched to the engine and exceeding the thrust load limits.



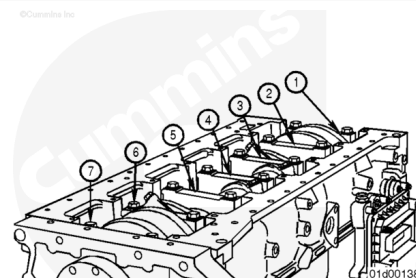
Remove

Before removing the main bearing caps, make sure that the caps are clearly marked for their location on the lubricating oil cooler side of the main bearing cap and cylinder block.

The number 1 cap is at the front of the engine.

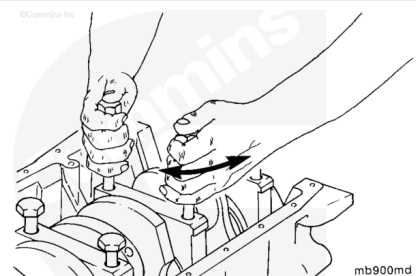


Note : For six-cylinder engines, replace number 2 through 6 bearings while the number 1 and number 7 caps support the crankshaft. After replacing number 2 through number 6, replace number 1 and number 7.



⚠ CAUTION ⚠

Do not pry on the main bearing caps to free them from the cylinder block. Damage to the main bearing caps and cylinder block can result.



Loosen the main bearing capscrews completely, but do **not** remove.

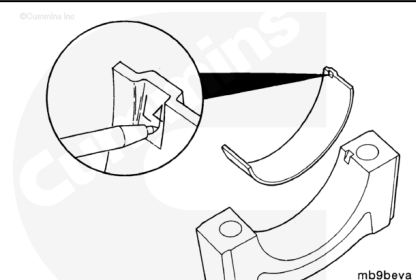
Use two of the main bearing cap bolts to “wiggle” the main bearing cap loose, being careful **not** to damage the bolt threads.

Remove the main bearing cap.

Mark the main bearings for position and number as they are removed.

Use an awl to mark the bearing's position in the tang area.

Note : Marking the bearing's position is for future identification or possible failure analysis.

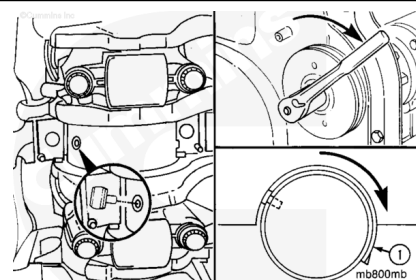


Follow this step to remove the upper main bearings, except for the number 1 front main bearing.

To remove the upper main bearing, install the Cummins® main bearing replacer, Part Number 3823818, in the oil hole of the crankshaft main bearing journal.

Use Cummins® barring tool, Part Number 3824591, to rotate the crankshaft so that the replacer contacts the upper main bearing on the side opposite the tang.

Continue to rotate the crankshaft in the direction that will remove the tang side (1) of the upper main bearing first.



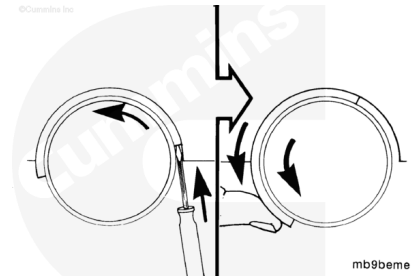
Remove the bearing.

⚠ CAUTION ⚠

Use care so the screwdriver does not damage the crankshaft or cylinder block.

Note : The front main bearing, number 1, does **not** have a hole in the journal, so the tool can **not** be used to replace the bearing.

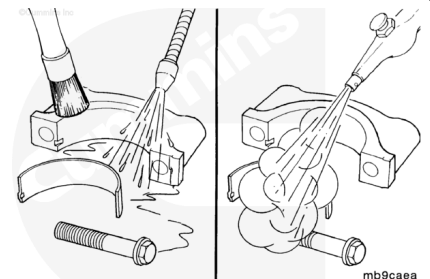
Use a flat blade screwdriver to gently bump the end of the bearing to loosen it from the cylinder block. Then, use finger pressure against the main bearing shell and rotate the crankshaft to roll the main bearing out.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

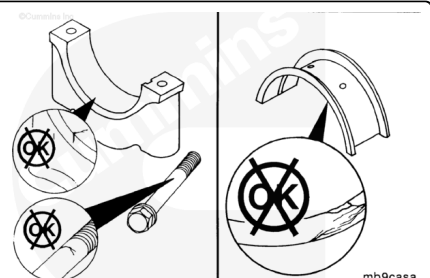
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Steam clean or use hot, soapy water to clean the main bearing caps.

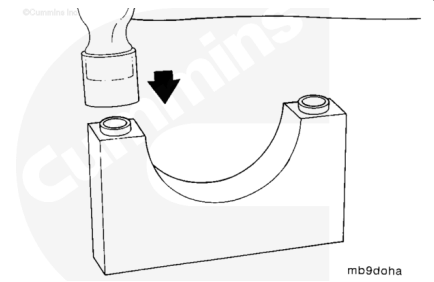
Dry with compressed air.

Inspect all main bearing caps, capscrews, and thrust bearings for deep scoring, overheating, etc.

Replace any damaged components. If the main bearing cap is damaged, the block **must** be replaced.



Check the main bearing caps to make sure the ring dowels are installed.



Check the crankshaft main bearing journals for damage or excessive wear. Minor scratches are acceptable.

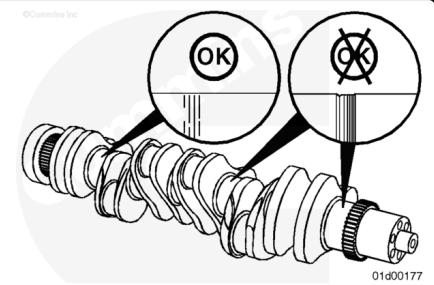
If crankshaft end clearance measured during the initial check was found to be out of specification, make sure to visually check the crankshaft thrust surface for excessive wear or damage. Minor scratches are acceptable.

Six-cylinder engines - the number 6 main bearing journal.

If damage is found, the crankshaft will need to be removed. Refer to Procedure 001-016 in Section 1.

(/qs3/pubsys2/xml/en/procedures/216/216-001-016-tr.html)

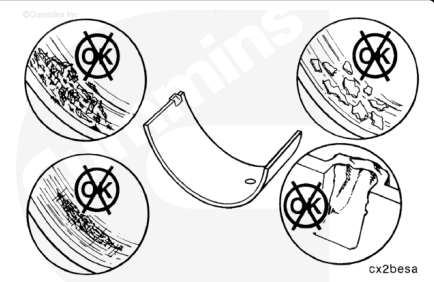
Also check the thrust bearing surfaces for excessive wear. Replace the thrust bearing(s), if excessive wear is found.



Inspect the bearings for damage.

Replace any bearings with the following damage:

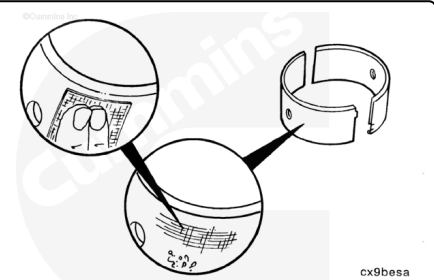
- Pitting
- Flaking
- Corrosion
- Lock tang damage
- Scratches.



Inspect the bearing shell seating surfaces for nicks or burrs. If nicks or burrs can **not** be removed with an abrasive pad, Part Number 3823258, or equivalent, the bearings **must** be replaced.

Note : If bearings are damaged they must be replaced as a set.

Note : For more detailed information on bearing damage, reference the "Analysis and Prevention of Bearing Failures" Bulletin 3810387.



Measure

Measure the main bearing shell thickness with an outside micrometer that has a ball tip.



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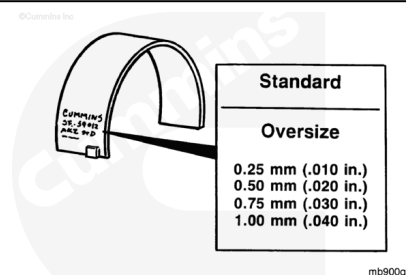
Main Bearing Dimensions			
	mm		in
Standard	2.446	MIN	0.0963
	2.464	MAX	0.0970
0.25 mm Oversize	2.571	MIN	0.1012
	2.589	MAX	0.1019
0.50 mm Oversize	2.696	MIN	0.1061
	2.714	MAX	0.1068
0.75 mm Oversize	2.821	MIN	0.1111
	2.839	MAX	0.1118
1.00 mm Oversize	2.946	MIN	0.1160
	2.964	MAX	0.1167

Discard a bearing shell if its thickness is below the minimum specifications.

Note : Main bearings are identified with a part number and size stamped on the back side.

If replacing the bearings, determine the size of the removed main bearings and obtain a set of the same size.

Oversize service main bearings are available for use with crankshafts that have been machined undersize. See the appropriate parts catalog.

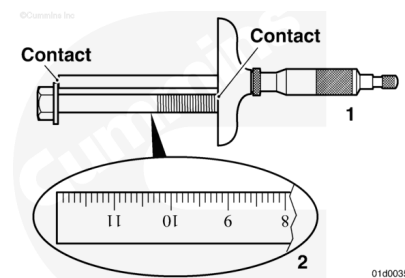


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Main Bearing Capscrew Reuse Measurement

CAUTION

This step must be completed. Failure to check the main bearing capscrew against reuse guidelines can result in severe engine damage.



To check if a main bearing capscrew can be reused, the length **must** be measured by performing the following:

For each main bearing capscrew that has been removed, measure the length from underneath the head of the capscrew to the tip of the capscrew, as illustrated, using one of two methods.

- A depth micrometer (preferred method for accuracy)
- A machinist's rule.

If the measurement is above the maximum specification, the main bearing capscrew **must** be replaced.

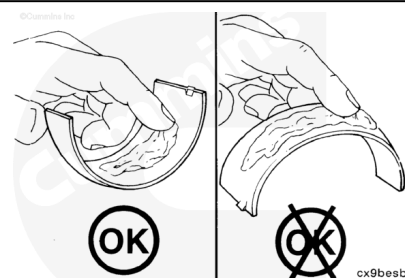
Main Bearing Underhead Capscrew Length		
mm		in
120.00	MAX	4.724

Install

Upper Main Bearing

⚠ CAUTION ⚠

Do not lubricate the side that is against the cylinder block.



Note : Make sure the main bearing being installed is the same size as the main bearing removed, if the crankshaft has **not** been machined. The size is engraved on the back of the main bearing.

Note : Install the upper main bearing cap after each upper main bearing is installed to keep the main bearing in place while the other upper main bearings are installed.

Apply assembly lube, Part Number 3163086, to the upper main bearings.

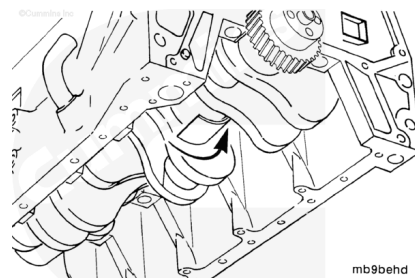
Note : The crankshaft thrust bearing must be installed in: For six-cylinder engines - the number 6 main bearing position.

Note : The upper and lower main bearing shells are not interchangeable. The backs of the main bearings are marked with the proper orientation.

Install the upper main bearings.

Insert the side of the main bearing opposite the tang first in between the crankshaft journal and block. Install the bearing as far as possible by hand.

When installing the thrust bearing, if necessary, push the crankshaft to the front or rear of the cylinder block.



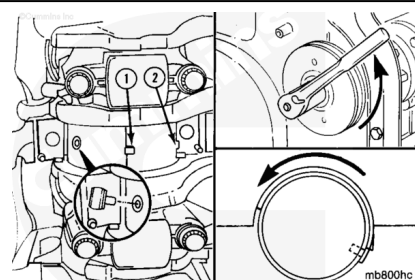
⚠ CAUTION ⚠

Make sure the pin does not slide under the bearing.

Follow this step to finish installing the upper main bearings, except for the number 1 front main bearing.

Use the Cummins® main bearing replacer, Part Number 3823818, to finish installing the main bearing. Rotate the crankshaft, using the Cummins® barring tool, Part Number 3824591.

Make sure the tang (1) on the main bearing is located in the notch (2) of the cylinder block. Finish pushing the main bearing into position.



⚠ CAUTION ⚠

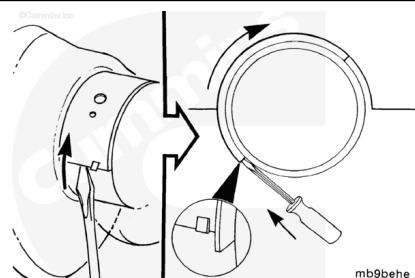
Use care so the screwdriver does not damage the crankshaft or cylinder block.

Note : The front main, number 1, does not have a hole in the journal so the pin can not be used to replace the bearing.

Install the number 1 main bearing.

Insert the side of the main bearing opposite the tang first and install as far as possible by hand.

Use a flat blade screwdriver to push the main bearing into position while rotating the crankshaft.

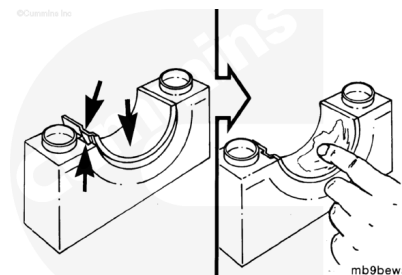


Make sure the tang on the main bearing is located in the notch of the cylinder block.

Lower Main Bearings

⚠ CAUTION ⚠

Do not lubricate the back side of the bearing that contacts the main bearing cap.



Make sure the backside of the bearings are clean and free of debris.

Install the lower main bearings into the main bearing caps.

Make sure to align the tangs of the bearings with tangs on the main bearing caps.

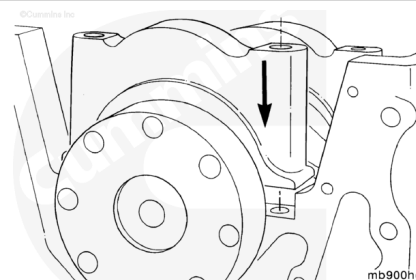
If equipped, install the lower crankshaft thrust bearing.

Six-cylinder engines - the number 6 main bearing position

Apply a coat of Cummins® assembly lube, Part Number 3163087, to the crankshaft side of the main bearings and thrust bearing surfaces.

⚠ CAUTION ⚠

Make sure the caps are correctly installed in the same position as removed, with the number towards the oil cooler side of the engine.

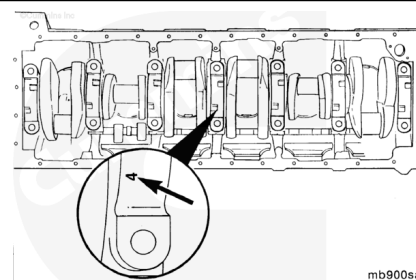


Make sure the main bearing cap surfaces between the main bearing cap and block are clean and free of debris.

Install the main bearing cap into position, aligning the main bearing cap dowel rings with the cylinder block.

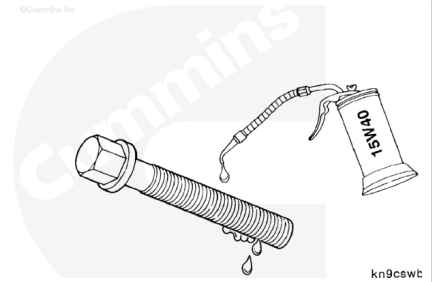
The main bearing caps are/were numbered during the removal process for their location. Number 1 starts with the front of the block.

Note : The caps must be installed so the numbers on the caps match the bearing saddle in the block. The lock tangs in the main bearing saddle and bearing cap must be on the same side.



Install the main bearing caps. Make sure to align the ring dowels on the main bearing cap with the corresponding drillings in the cylinder block.

Lubricate the main bearing capscrew threads and underside of the head with clean engine oil.

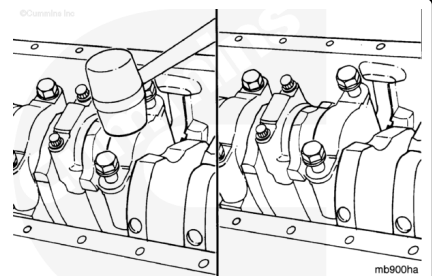


Gently tap the main bearing cap into position with a plastic or rubber mallet.

When seated, install the main bearing capscrews and tighten.

Torque Value: 50 n·m [37 ft-lb]

Do **not** tighten to the final torque value at this time. Final torque **must** be applied after all main bearing caps are installed.

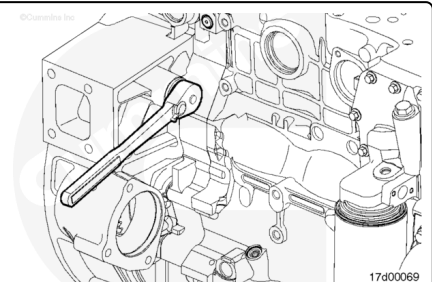


Use the Cummins® barring tool, Part Number 3824591. The crankshaft **must** rotate freely after installing the main bearing caps.

While applying final torque to the main bearing capscrews, frequently check that the crankshaft rotates freely.

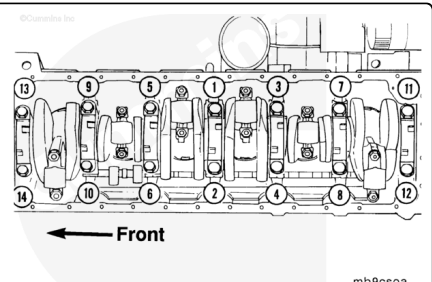
If the crankshaft does **not** rotate freely:

- Check if the crankshaft is contacting one of the connecting rods
- Check if the correct main bearing caps were installed correctly
- Check if the main bearing cap ring dowels or mounting surfaces were damaged during installation
- Check if the correct main bearings were installed.



⚠ CAUTION ⚠

There is a different torque procedure for new and previously installed main bearing capscrews. Failure to use the correct torque process can result in engine damage.



Note : The sequence to the right is for a six cylinder engine. For a four cylinder engine, use the same sequence for the five main bearing caps.

Tighten the main bearing capscrews evenly and in sequence.

Torque Value:

New Main Bearing Capscrews

1. 120 n•m [89 ft-lb]
2. Loosen completely
3. 60 n•m [44 ft-lb]
4. 85 n•m [63 ft-lb]
5. Rotate 120 degrees.

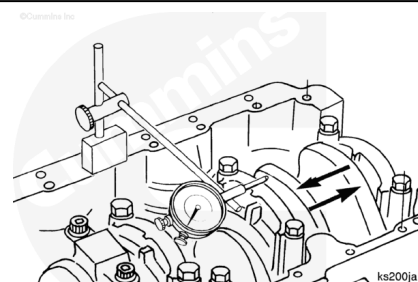
Torque Value:

Previously Installed Main Bearing Capscrews

1. 60 n•m [44 ft-lb]
2. 80 n•m [59 ft-lb]
3. Rotate 90 degrees.

Note : The dimensions of the thrust bearing and crankshaft journal determine end clearance.

Measure the crankshaft end clearance using Cummins® dial indicator, Part Number 3824564, and Cummins® magnetic base, Part Number 3377399.



Crankshaft End Clearance		
mm		in
0.102	MIN	0.004
0.432	MAX	0.017

If the crankshaft end clearance is **not** within specification:

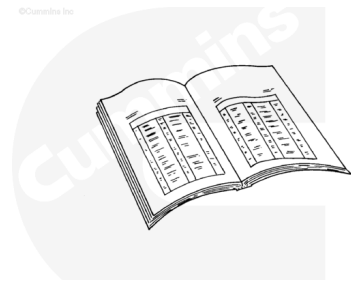
- If the crankshaft end clearance is below specification, check if there are any obstructions limiting the crankshaft's travel (lubricating oil pump, connecting rod, etc.)
- If the crankshaft end clearance is above specification, inspect the crankshaft thrust bearing surface. Also check if the correct thrust bearing(s) were installed.

Note : Oversize thrust bearings are available if the end clearance is **not** within specifications. Oversize thrust bearings of 0.25 to 0.51 mm [0.010 to 0.020 in] are available.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Install block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/220/220-001-089-tr.html>)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/220/220-007-035-tr.html>)
- Install the lubricating oil pan and gasket. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/216/216-007-025-tr.html>)
- Fill the lubricating oil pan. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/216/216-007-037.html>)
- Connect the batteries. Refer to the equipment service information.
- Operate the engine at idle for 5 to 10 minutes. Check for loose parts, leaks, and proper oil pressure.